

# Cloud Computing

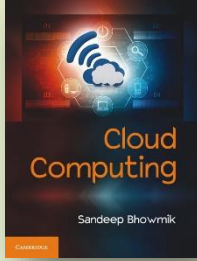
Sandeep Bhowmik

## Chapter 4



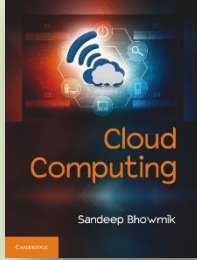
### Cloud Computing Model (Contd.)

Cambridge University Press



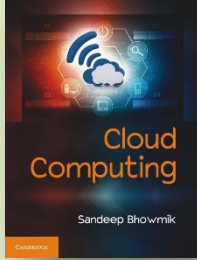
## Cloud Deployment Model

- Cloud services can be arranged or deployed in a number of ways.
- The deployment choice depends on the requirements of the consumer organization.



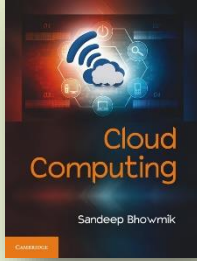
## Cloud Deployment Model

- The deployment model describes -
  - the utility of a cloud
  - specifies its access boundary
  - indicates the relative location of the cloud with respect to the location of consumer



## Cloud Deployment Model

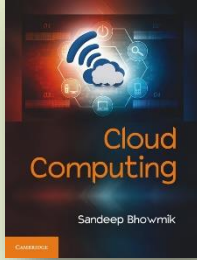
- **Public Cloud**
- Physical location wise this cloud remains external or off-premises.
- Consumers remotely access the service.
- Public cloud is also referred as *external cloud*.
- Public cloud shares one-to-many relationship with consumers.



## Cloud Deployment Model

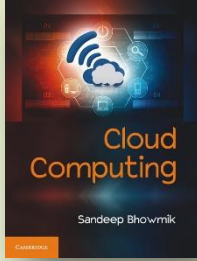
### ➤ Public Cloud

- It is hosted and managed by some computing vendor, who establishes datacenters to provide service.
- Consumers under this cloud deployment model are entirely free from any headache of infrastructure administration and system management.
- Consumers have low degree of control over the cloud.



## Cloud Deployment Model

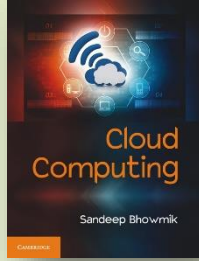
- **Public Cloud**
- Public cloud deployment promotes multi-tenancy at its highest degree.
- This increases resource utilization rates and decreases vendor's cost of service delivery.
- Through this model consumers can access potentially superior service at a lower cost.
- Amazon Web Services, Google Cloud, Microsoft Azure and Salesforce.com are popular public clouds.



## Cloud Deployment Model

- **Private Cloud**
- This deployment does not provide open access to all.
- It is mainly for organizational use, and access to a private cloud deployment is restricted for general public.
- Private cloud is also referred as *internal cloud*.



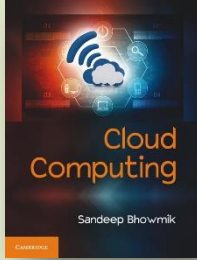


## Cloud Deployment Model

### ➤ Private Cloud

- Private cloud is user for high security and critical systems.
- Private cloud may reside either inside consumer organization's premises (on-premises) or outside (off-premises) at any neutral location.
- private cloud shares one-to-one relationship with consumer.





## Cloud Deployment Model

### ➤ Private Cloud

- The feature of multi-tenancy does not apply in private cloud.
- Consumers have more control over the cloud.
- The cost of availing private cloud is higher than public cloud.

## Cloud Deployment Model

### Private Cloud vs. Public Cloud

| Private Cloud                                                                | Public Cloud                                                 |
|------------------------------------------------------------------------------|--------------------------------------------------------------|
| Can be both of type on-premises or off-premises.                             | There can't be any on-premises public cloud deployment.      |
| On-premises private cloud can be delivered over private network.             | Can only be delivered over public network.                   |
| Does not support multi-tenancy for unrelated and external tenants.           | Demonstrates multi-tenancy capability with its full ability. |
| Resources are for exclusive use of one consumer (generally an organization). | Resources are shared among multiple consumers.               |
| A private cloud is accessible to a restricted number of people.              | This facility is accessible to anyone.                       |

## Cloud Deployment Model

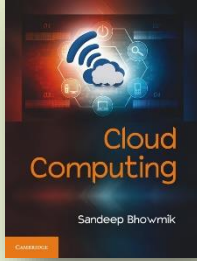
### Private Cloud vs. Public Cloud

| Private Cloud                                                                                                               | Public Cloud                                                                                  |
|-----------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| This is for organizational use.                                                                                             | Can be used both by organization and user.                                                    |
| Consumers have important roles to play in management of the cloud.                                                          | Management is entirely provider's responsibility.                                             |
| Consumers have more control over the environment.                                                                           | Consumers have very less control or no control.                                               |
| Provides more confidence regarding data security as remains under the control of consumer organization's security boundary. | Public cloud deployment often creates concerns regarding security and privacy of data.        |
| Cost of computing is more in comparison to public cloud.                                                                    | This is more economical, as multiple unrelated consumers (tenants) share same infrastructure. |

## Cloud Deployment Model

### Private Cloud vs. Public Cloud

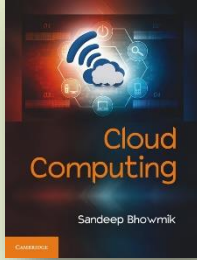
| Private Cloud                                               | Public Cloud                                                                    |
|-------------------------------------------------------------|---------------------------------------------------------------------------------|
| Not ideal for promoting pay-as-you-use philosophy.          | Ideal model for practicing pay-as-you-use philosophy.                           |
| Resource may often remain idle, resulting resource wastage. | Resource utilization is optimum, due to presence of larger number of consumers. |
| Not so environment friendly like public cloud.              | Promotes green computing at its best.                                           |



## Cloud Deployment Model

### ➤ Community Cloud

- This deployment model allows access to a number of organizations or consumers - belonging to a community.
- The model is built to serve some common and specific purpose.
- This model allows sharing of infrastructure and resources among multiple consumers belonging to a single community.



## Cloud Deployment Model

- **Community Cloud**
- Community cloud deployment can be on-premises or off-premises.
- It may be identified as a generalized form of private cloud.
- It is cheaper compared to private cloud.
- Community cloud provides advantages of public cloud deployment, combined with the promise of private cloud like security and privacy of data.

## Cloud Deployment Model

### Private Cloud vs. Community Cloud

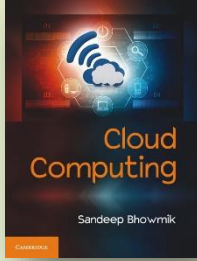
| Private Cloud                                                           | Community Cloud                                                      |
|-------------------------------------------------------------------------|----------------------------------------------------------------------|
| One private cloud can be used by one consumer (may be an organization). | One community cloud can be used by one community.                    |
| Access is restricted among members of a single consumer.                | Access is restricted among members of a single community.            |
| Resources of a private cloud are for exclusive use of one consumer.     | Resources are shared among multiple consumers of a single community. |
| May reside at consumer's premises.                                      | May reside at some consumer's premises within the community.         |



## Cloud Deployment Model

### Private Cloud vs. Community Cloud

| Private Cloud                                                      | Community Cloud                                                                          |
|--------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Private cloud (on-premises) can be delivered over private network. | To provide access to multiple consumers (of a community), delivered over public network. |
| Does not support multi-tenancy.                                    | Multi-tenancy is supported.                                                              |
| This is for organizational use.                                    | This one is for use of communities.                                                      |



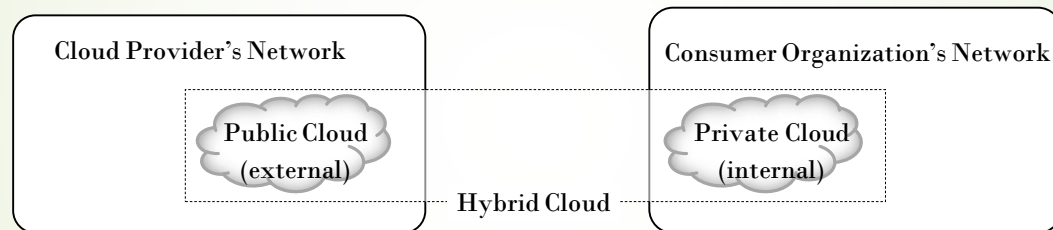
## Cloud Deployment Model

### ➤ Hybrid Cloud

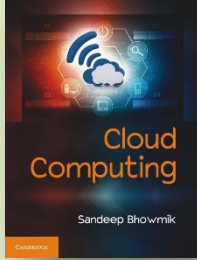
- It is created by combining private or community deployment with public cloud deployment together.
- This model helps businesses to take advantage of private or community cloud by storing critical applications and data there.
- It also provides cost benefit by keeping shared data and applications on the public cloud.

# Cloud Deployment Model

## ➤ Hybrid Cloud



A hybrid cloud model combining public cloud with on-premises private cloud

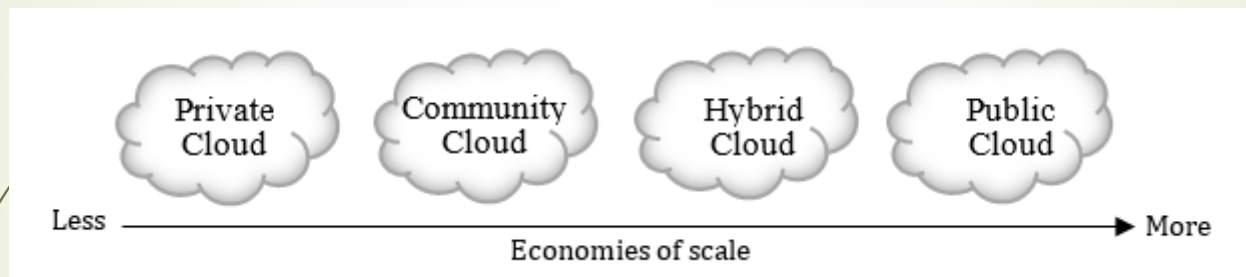


## Cloud Deployment Model

- Cloud deployment states how exclusively a cloud consumer can use computing resources.
- To select the appropriate cloud deployment model, one will have to
  - assess the business needs,
  - strengths and weaknesses of a model,
  - privacy/security requirements, and
  - goals of the organization.

## Cloud Deployment Model

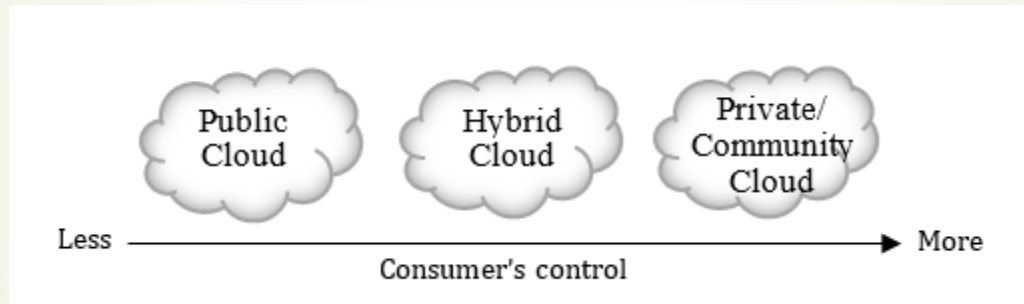
### ➤ Economies of Scale



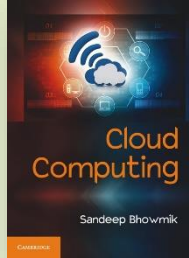
Variation of cost-effectiveness with different cloud deployments

## Cloud Deployment Model

### ➤ Consumer's Authority



Variation in user's control over different cloud deployments



**Thank You**